Pf post lab task 10

Q1:

```
#include <stdio.h>


int main() {


   int attendance[2][3][4];
   int i, j, k;



   printf("Enter attendance (1 for present, 0 for absent):\n");
   for (i = 0; i < 2; i++) {
      for (j = 0; j < 3; j++) {
         printf("Enter 4 days attendance for Class %d, Student %d: ", i + 1, j + 1);
         for (k = 0; k < 4; k++) {
            scanf("%d", &attendance[i][j][k]);
         }
      }
   }


   int class1_total = 0, class2_total = 0;
   int perfect_attendance_count = 0;



   printf("\n--- RESULTS ---\n");
```

```
for (i = 0; i < 2; i++) {
    printf("\nClass %d:\n", i + 1);
    int max_days = -1;
    int best_student = 0;

    for (j = 0; j < 3; j++) {
        int present_days = 0;


        for (k = 0; k < 4; k++) {
            present_days += attendance[i][j][k];
        }


        printf("  Student %d was present for %d days\n", j + 1, present_days);


        if (i == 0) class1_total += present_days;
        else class2_total += present_days;

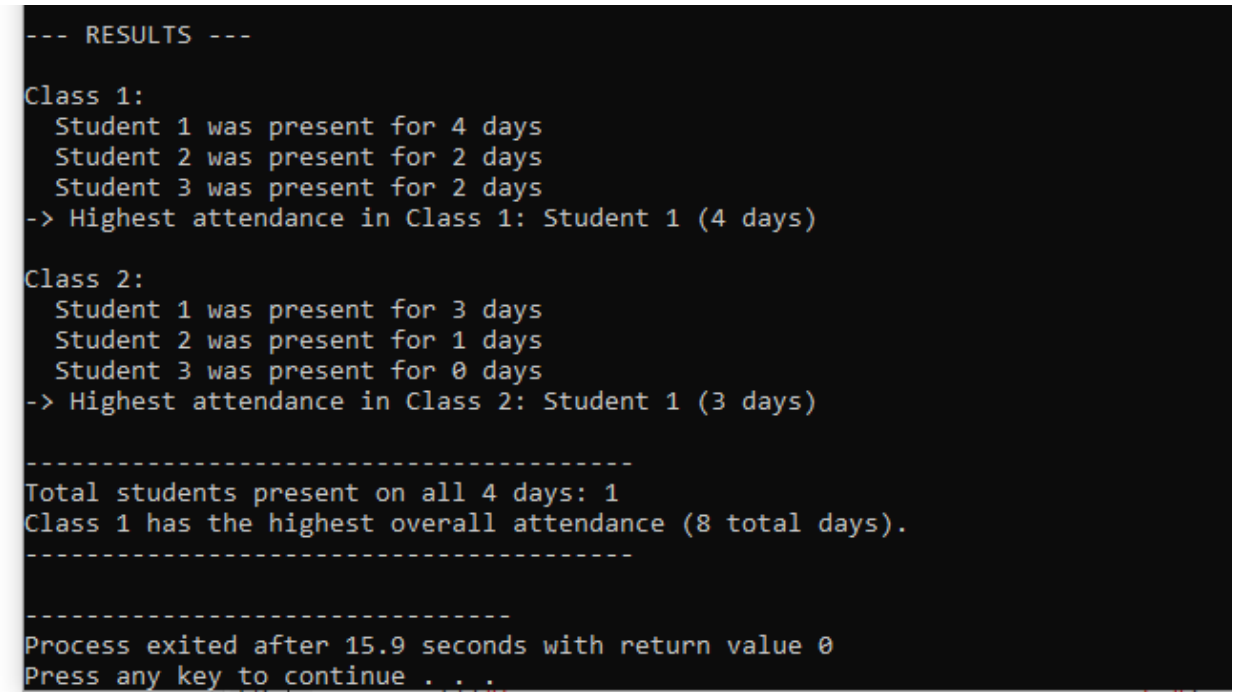

        if (present_days > max_days) {
            max_days = present_days;
            best_student = j + 1;
        }
```

```
            if (present_days == 4) {
                perfect_attendance_count++;
            }
        }
        printf("-> Highest attendance in Class %d: Student %d (%d days)\n", i + 1, best_student,
max_days);
    }


    printf("\n----------------------------------------\n");
    printf("Total students present on all 4 days: %d\n", perfect_attendance_count);


    if (class1_total > class2_total) {
        printf("Class 1 has the highest overall attendance (%d total days).\n", class1_total);
    } else if (class2_total > class1_total) {
        printf("Class 2 has the highest overall attendance (%d total days).\n", class2_total);
    } else {
        printf("Both classes have equal overall attendance.\n");
    }
    printf("----------------------------------------\n");

    return 0;
}
```



Q2:

```
#include <stdio.h>


int main() {
   int a = 10;
   int b = 20;


   int *ptr1 = &a;
   int *ptr2 = &b;



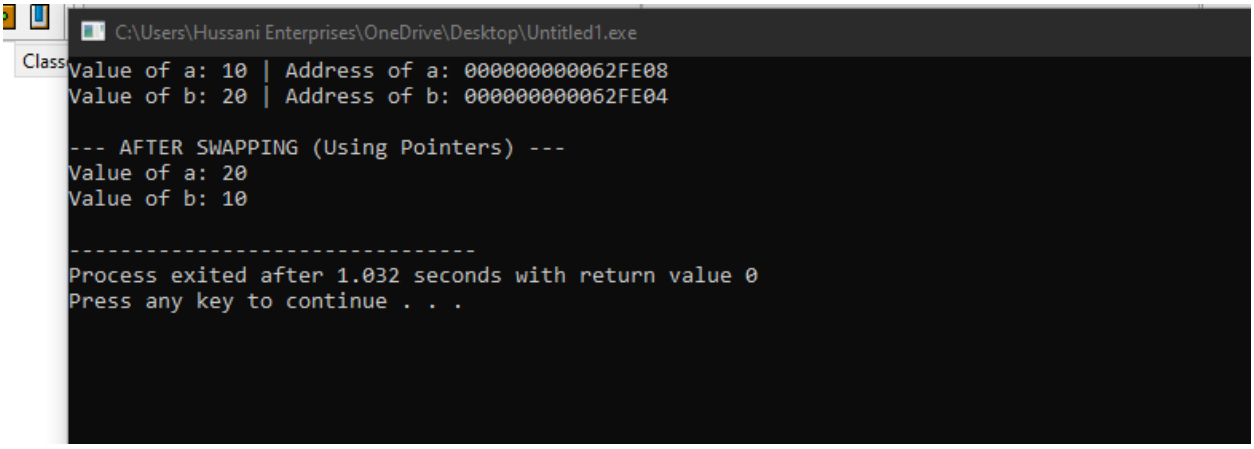
   printf("Value of a: %d | Address of a: %p\n", *ptr1, ptr1);
   printf("Value of b: %d | Address of b: %p\n", *ptr2, ptr2);
```

```
    int temp = *ptr1;
    *ptr1 = *ptr2;
    *ptr2 = temp;



    printf("\n--- AFTER SWAPPING (Using Pointers) ---\n");
    printf("Value of a: %d\n", *ptr1);
    printf("Value of b: %d\n", *ptr2);


    return 0;
}
```



Q3:

```
#include <stdio.h>

int main() {

   int num1, num2;



   int *ptr1, *ptr2;



   printf("Enter first integer: ");
   scanf("%d", &num1);

   printf("Enter second integer: ");
   scanf("%d", &num2);



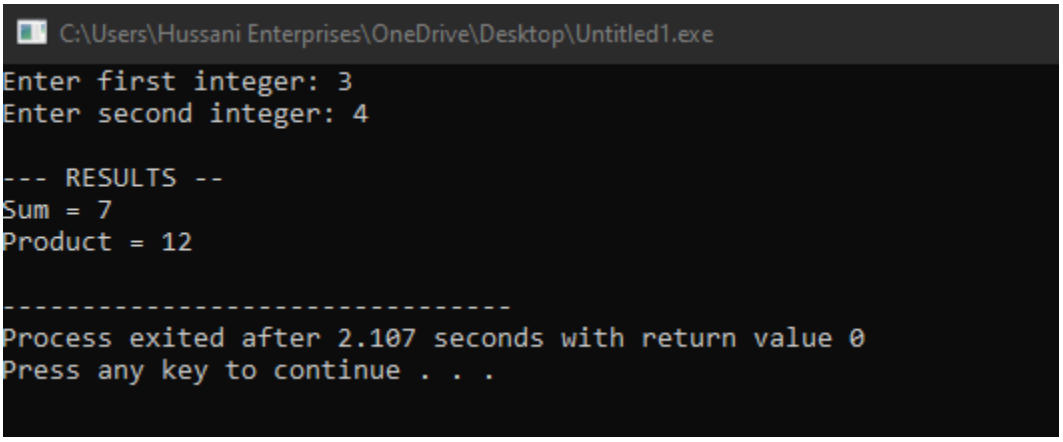
   ptr1 = &num1;
   ptr2 = &num2;



   int sum = *ptr1 + *ptr2;
   int product = (*ptr1) * (*ptr2);
```

```
    printf("\n--- RESULTS --\n");
    printf("Sum = %d\n", sum);
    printf("Product = %d\n", product);

    return 0;
}
```



Q4:

```
#include <stdio.h>

// 1. Declare the user-defined function for calculating the bill
float calculateBill(int units, char category) {
    float bill = 0;

    // Calculate base bill based on customer category
    if (category == 'D' || category == 'd') {
        // Domestic: Rs. 15 for first 100 units, Rs. 20 for additional units[cite: 4, 5]
        if (units <= 100) {
            bill = units * 15;
        } else {
```

```
            bill = (100 * 15) + ((units - 100) * 20);
        }
    }
    else if (category == 'C' || category == 'c') {
        // Commercial: Rs. 25 per unit for all units[cite: 4, 5]
        bill = units * 25;
    }

    return bill; // Return the calculated base bill back to main
}

int main() {
    int units;
    char category;
    float bill = 0, surcharge = 0, final_amount = 0;


    printf("Enter customer category ('D' for Domestic, 'C' for Commercial): ");
    scanf(" %c", &category);

    printf("Enter units consumed: ");
    scanf("%d", &units);


    if (category != 'D' && category != 'd' && category != 'C' && category != 'c') {
        printf("Invalid category entered!\n");
```

```
        return 0;
    }

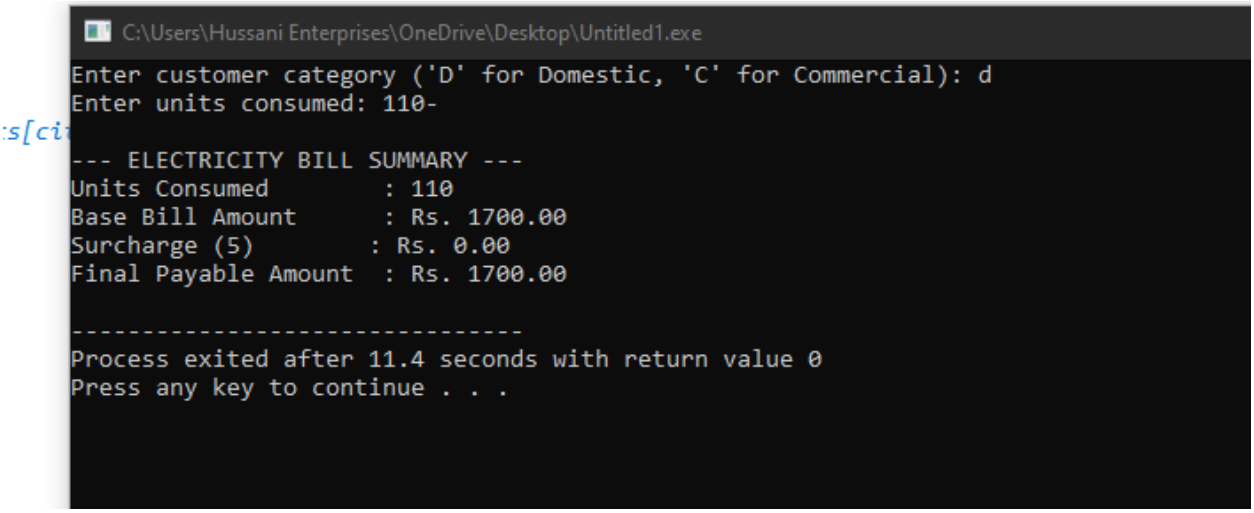

    bill = calculateBill(units, category);


    if (bill > 5000) {
        surcharge = bill * 0.05;
    }


    final_amount = bill + surcharge;

    printf("\n--- ELECTRICITY BILL SUMMARY ---\n");
    printf("Units Consumed       : %d\n", units);
    printf("Base Bill Amount     : Rs. %.2f\n", bill);
    printf("Surcharge (5%)       : Rs. %.2f\n", surcharge);
    printf("Final Payable Amount  : Rs. %.2f\n", final_amount);

    return 0;
}
```



Q5:

```
#include <stdio.h>



float calculateBonus(float basic_salary, int years) {
    float bonus = 0;



    if (years > 5) {
        bonus = basic_salary * 0.15;
    }


    else if (years >= 3 && years <= 5) {
        bonus = basic_salary * 0.10;
    }
```

```
    else {
        bonus = basic_salary * 0.05;
    }

    return bonus;
}

int main() {
    float basic_salary, bonus = 0, tax = 0, salary_after_bonus = 0, final_salary = 0;
    int years;


    printf("Enter basic salary: ");
    scanf("%f", &basic_salary);

    printf("Enter years of service: ");
    scanf("%d", &years);


    bonus = calculateBonus(basic_salary, years);


    salary_after_bonus = basic_salary + bonus;

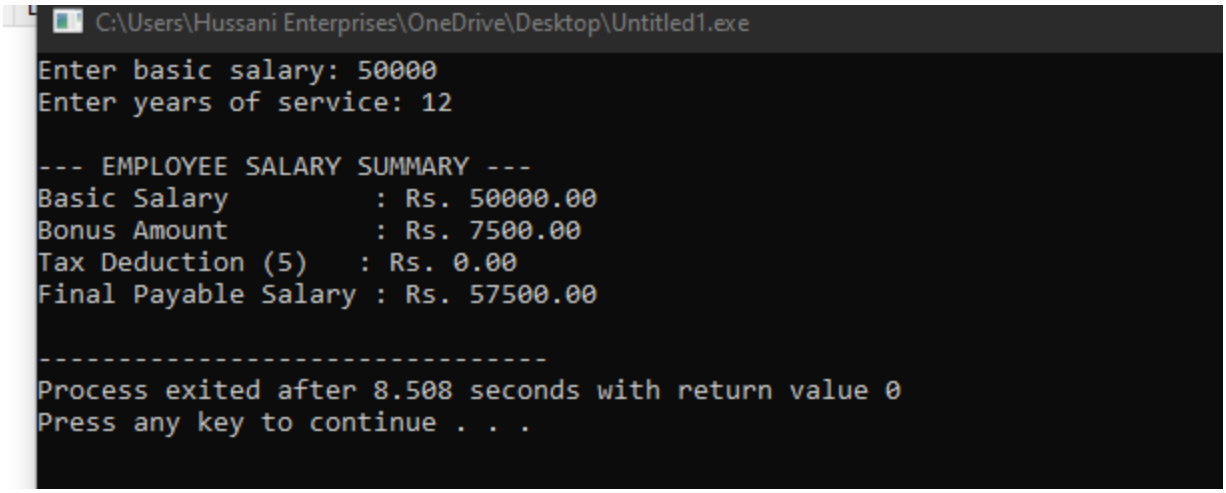

    if (salary_after_bonus > 100000) {
```

```
    tax = salary_after_bonus * 0.05;
  }


  final_salary = salary_after_bonus - tax;

  printf("\n--- EMPLOYEE SALARY SUMMARY ---\n");
  printf("Basic Salary        : Rs. %.2f\n", basic_salary);
  printf("Bonus Amount        : Rs. %.2f\n", bonus);
  printf("Tax Deduction (5%)   : Rs. %.2f\n", tax);
  printf("Final Payable Salary : Rs. %.2f\n", final_salary);

  return 0;
}
```



Q6;

```
#include <stdio.h>
```

```
float calculateRemainingIncome(float salary, float existing_payments) {
    return salary - existing_payments;
}

int main() {
    int age;
    float salary, existing_payments, remaining_income, max_loan;
    int is_eligible = 1;

    printf("Enter age: ");
    scanf("%d", &age);

    printf("Enter monthly salary: ");
    scanf("%f", &salary);

    printf("Enter existing monthly loan payments: ");
    scanf("%f", &existing_payments);

    printf("\n--- LOAN ELIGIBILITY REPORT ---\n");

    if (age < 21) {
        printf("Ineligible: Age must be at least 21 years.\n");
        is_eligible = 0;
    }

    if (salary < 50000) {
```

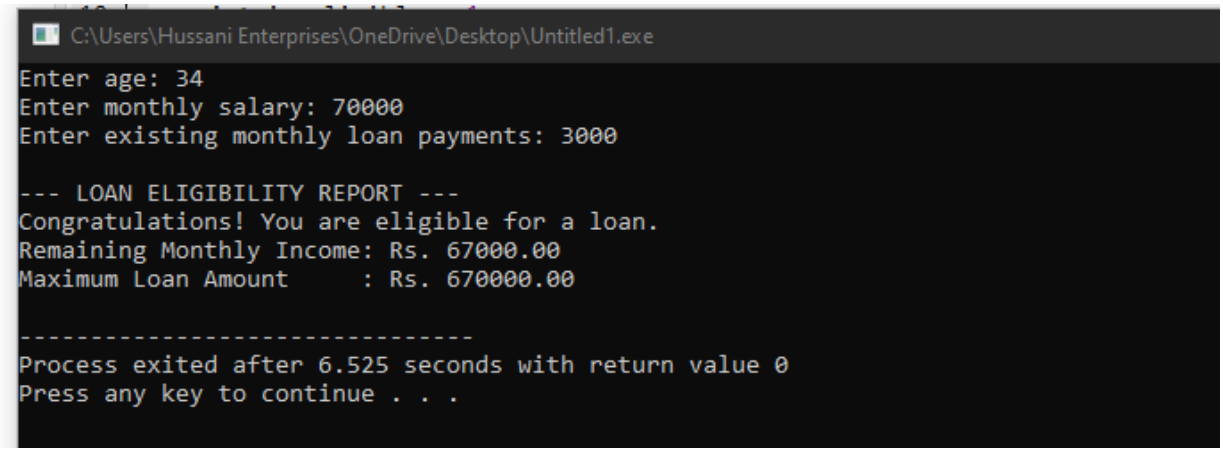
```
        printf("Ineligible: Monthly salary must be at least Rs. 50,000.\n");
        is_eligible = 0;
    }

    remaining_income = calculateRemainingIncome(salary, existing_payments);

    if (remaining_income < (salary * 0.40)) {
        printf("Ineligible: Must have at least 40%% of salary remaining after existing payments.\n");
        is_eligible = 0;
    }

    if (is_eligible == 1) {
        max_loan = remaining_income * 10;
        printf("Congratulations! You are eligible for a loan.\n");
        printf("Remaining Monthly Income: Rs. %.2f\n", remaining_income);
        printf("Maximum Loan Amount     : Rs. %.2f\n", max_loan);
    }

    return 0;
}
```



Q7":

```
#include <stdio.h>


int checkPalindrome(int n) {
   int reversed = 0, remainder, original = n;
   while (original != 0) {
      remainder = original % 10;
      reversed = reversed * 10 + remainder;
      original /= 10;
   }
   if (n == reversed) return 1;
   else return 0;
}


int checkPrime(int n) {
   if (n <= 1) return 0;
   for (int i = 2; i <= n / 2; i++) {
      if (n % i == 0) return 0;
```

```
    }
    return 1;
}

int main() {
    int num;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

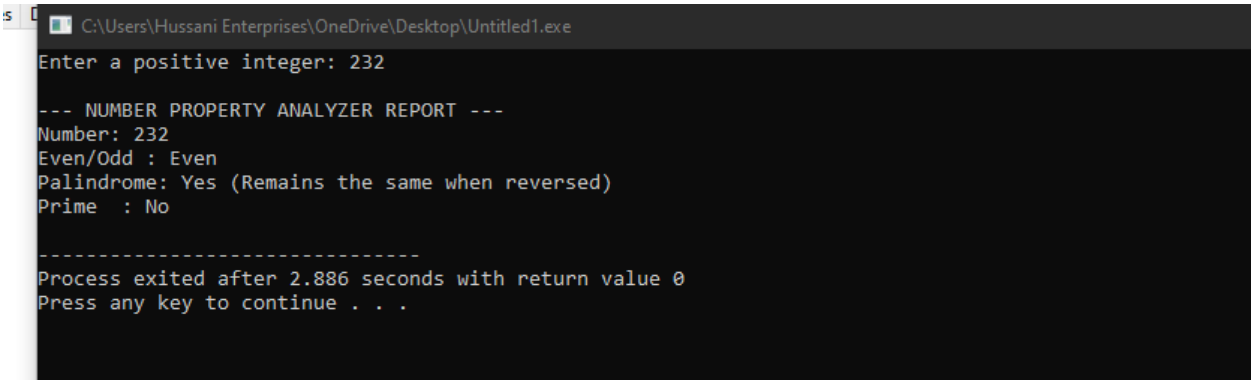
    printf("\n--- NUMBER PROPERTY ANALYZER REPORT ---\n");
    printf("Number: %d\n", num);


    if (num % 2 == 0) {
        printf("Even/Odd : Even\n");
    } else {
        printf("Even/Odd : Odd\n");
    }
```

```
    if (checkPalindrome(num)) {
        printf("Palindrome: Yes (Remains the same when reversed)\n");
    } else {
        printf("Palindrome: No\n");
    }


    if (checkPrime(num)) {
        printf("Prime  : Yes\n");
    } else {
        printf("Prime  : No\n");
    }

    return 0;
}
```



Q8;

```
#include <stdio.h>


float calculateBaseCharges(float weight) {
   if (weight <= 2) {
      return 200;
   } else if (weight <= 5) {
      return 400;
   } else {
      return 700;
   }
}


float calculateDistanceCharges(float distance) {
   if (distance > 10) {
      return (distance - 10) * 20;
   }
   return 0;
}


int main() {
   float weight, distance, base_charges = 0, distance_charges = 0, total_charges = 0, discount =
0, final_amount = 0;
   char is_member;


   printf("Enter parcel weight (in kg): ");
```

```
    scanf("%f", &weight);

    printf("Enter delivery distance (in km): ");
    scanf("%f", &distance);

    printf("Is the customer a member? (y/n): ");
    scanf(" %c", &is_member);

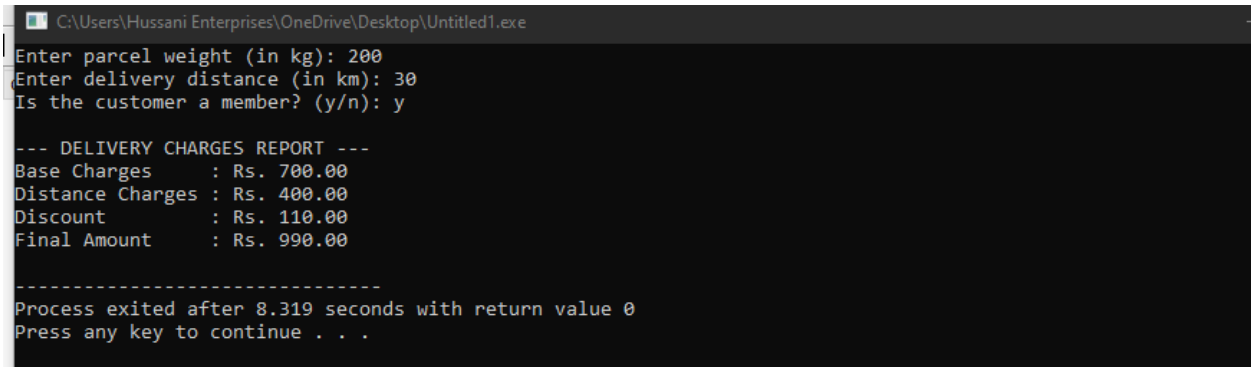
    base_charges = calculateBaseCharges(weight);
    distance_charges = calculateDistanceCharges(distance);
    total_charges = base_charges + distance_charges;

    if ((is_member == 'y' || is_member == 'Y') && total_charges > 1000) {
        discount = total_charges * 0.10;
    }

    final_amount = total_charges - discount;

    printf("\n--- DELIVERY CHARGES REPORT ---\n");
    printf("Base Charges     : Rs. %.2f\n", base_charges);
    printf("Distance Charges : Rs. %.2f\n", distance_charges);
    printf("Discount         : Rs. %.2f\n", discount);
    printf("Final Amount     : Rs. %.2f\n", final_amount);

    return 0;
}
```



Q9:

```
#include <stdio.h>


void checkBalance(float balance) {
   printf("Current Balance: Rs. %.2f\n", balance);
}


float depositMoney(float balance) {
   float amount;
   printf("Enter amount to deposit: ");
   scanf("%f", &amount);


   if (amount <= 0) {
      printf("Error: Deposit amount must be positive.\n");
   } else {
      balance += amount;
      printf("Successfully deposited Rs. %.2f\n", amount);
      printf("Updated Balance: Rs. %.2f\n", balance);
   }
   return balance;
```

```
}

float withdrawMoney(float balance) {
   float amount;
   printf("Enter amount to withdraw: ");
   scanf("%f", &amount);

   if (amount <= 0) {
      printf("Error: Withdrawal amount must be positive.\n");
   } else if (amount > 10000) {
      printf("Error: Withdrawal cannot exceed Rs. 10,000 per transaction.\n");
   } else if (amount > balance) {
      printf("Error: Withdrawal cannot exceed the available balance.\n");
   } else {
      balance -= amount;
      printf("Successfully withdrew Rs. %.2f\n", amount);
      printf("Updated Balance: Rs. %.2f\n", balance);

      if (balance < 5000) {
         printf("WARNING: Your balance has fallen below Rs. 5,000!\n");
      }
   }
   return balance;
}

int main() {
```

```
float balance = 20000.0;
int choice;

do {
   printf("\n--- ATM TRANSACTION SYSTEM ---\n");
   printf("1. Check Balance\n");
   printf("2. Deposit Money\n");
   printf("3. Withdraw Money\n");
   printf("4. Exit\n");
   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch (choice) {
      case 1:
         checkBalance(balance);
         break;
      case 2:
         balance = depositMoney(balance);
         break;
      case 3:
         balance = withdrawMoney(balance);
         break;
      case 4:
         printf("Thank you for using the ATM. Goodbye!\n");
         break;
      default:
```

```
            printf("Invalid choice! Please choose between 1 and 4.\n");
        }
    } while (choice != 4);

    return 0;
}
```

